

COMPANION SLIDES

Chapter 1

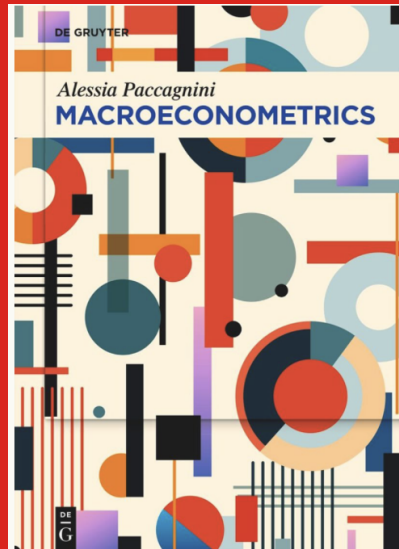
Univariate Time Series

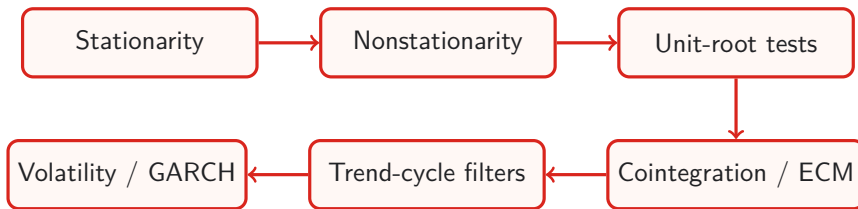
From Stationarity to Cointegration

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Key takeaway

Before estimating dynamics, ask what kind of persistence is present in the data.

- ▶ Distinguish strong and weak stationarity.
- ▶ Recognize AR, MA and ARMA dynamics.
- ▶ Use ACF and PACF for identification.
- ▶ Understand Wold decomposition.
- ▶ Separate deterministic from stochastic trends.
- ▶ Apply unit-root tests appropriately.
- ▶ Diagnose spurious regression.
- ▶ Understand cointegration and error correction.
- ▶ Compare trend-cycle decompositions.
- ▶ Model volatility clustering with ARCH/GARCH.

Weak stationarity

A process y_t is weakly stationary when $E(y_t) = \mu$, $Var(y_t) = \gamma_0 < \infty$, and $Cov(y_t, y_{t-k}) = \gamma_k$ depends on the lag k , not on calendar time.

Why it matters

- ▶ Stable probabilistic structure
- ▶ Meaningful autocovariances
- ▶ Standard time-series asymptotics become available

White noise

$\varepsilon_t \sim WN(0, \sigma^2)$ with zero serial correlation is the basic innovation from which many stationary models are built.

AR(p)

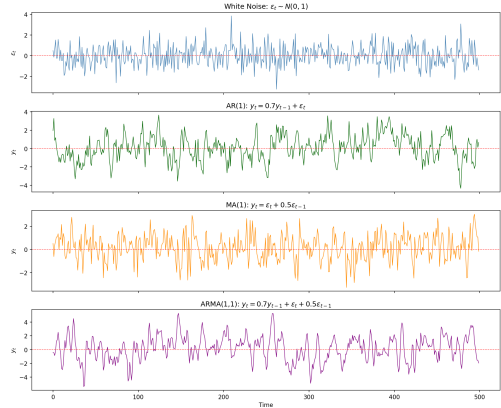
$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

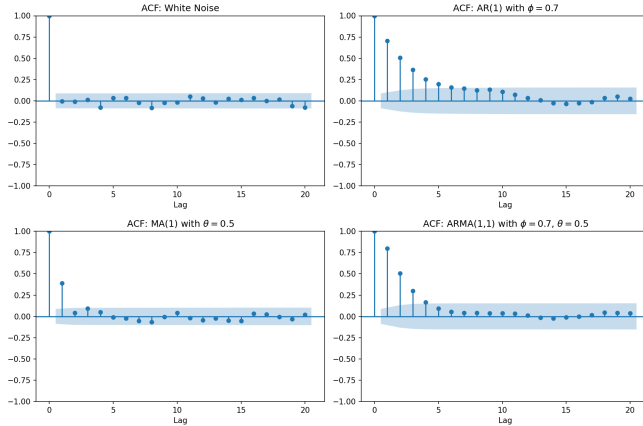
MA(q)

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

ARMA(p, q)

Combines autoregressive persistence with moving-average shock dynamics.





AR processes typically show a gradually decaying ACF; MA processes have an ACF that cuts off; ARMA processes usually show decay in both ACF and PACF.

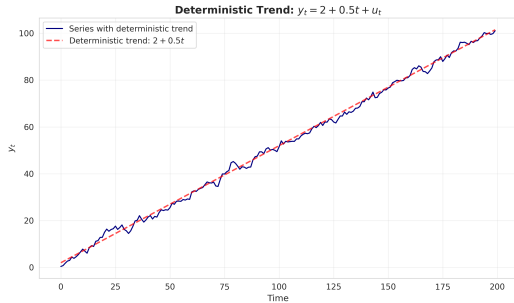
Wold's theorem

Every covariance-stationary, purely nondeterministic process can be represented as an infinite moving average:

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1, \quad \sum_j \psi_j^2 < \infty.$$

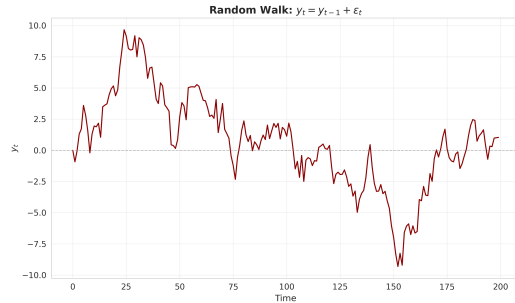
- ▶ Shocks ε_t are the new information arriving at time t .
- ▶ The coefficients ψ_j describe how shocks propagate through time.
- ▶ This representation is the conceptual bridge to impulse-response analysis later in the book.

Deterministic trend

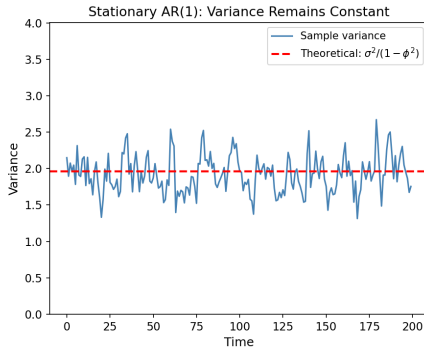
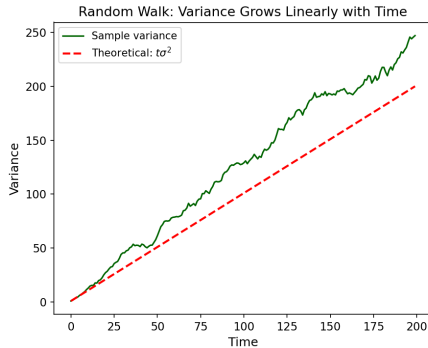


Shocks are temporary around a deterministic path.

Stochastic trend



A unit-root shock has a permanent effect on the level.



Key takeaway

Trend-stationary and difference-stationary processes require different transformations and imply different responses to shocks.

If y_t becomes stationary after differencing d times, an $ARIMA(p, d, q)$ model satisfies

$$\phi(L)(1 - L)^d y_t = c + \theta(L)\varepsilon_t.$$

AR

persistence in the
stationary component

I

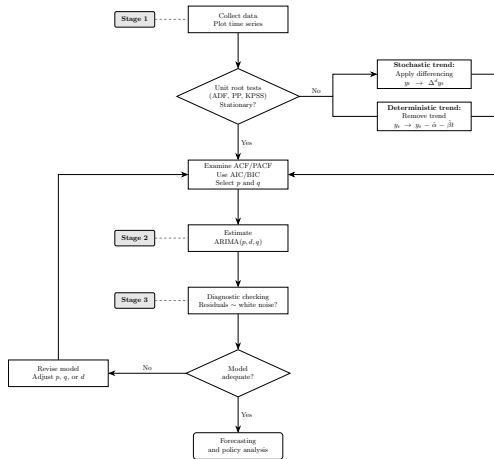
order of differencing
needed for stationarity

MA

dynamics of innovations

Key takeaway

Do not difference mechanically: first understand the source of nonstationarity.



ADF / PP

H_0 : unit root

H_1 : stationarity

- ▶ Dickey-Fuller and Augmented Dickey-Fuller
- ▶ Phillips-Perron: robust treatment of serial correlation / heteroskedasticity

KPSS

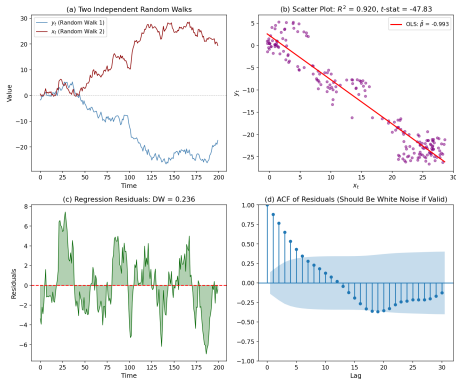
H_0 : stationarity

H_1 : unit-root-type nonstationarity

- ▶ Useful complement to ADF/PP
- ▶ Results depend on deterministic terms and finite-sample power

Key takeaway

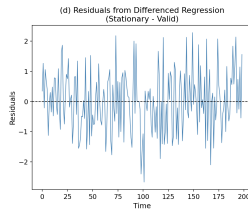
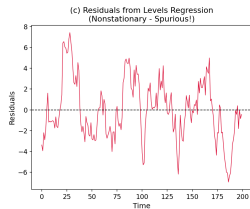
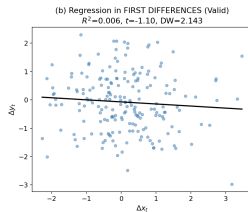
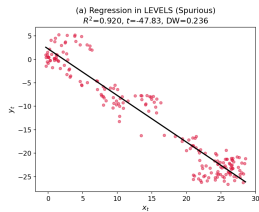
Use complementary evidence rather than treating one test as an automatic decision rule.



Two unrelated persistent series can generate:

- ▶ high R^2
- ▶ apparently significant t -statistics
- ▶ persistent residuals
- ▶ misleading economic conclusions

Persistence can masquerade as explanatory power.



If variables are nonstationary, ask whether they should be differenced or whether a stationary long-run combination exists.

Suppose x_t and y_t are both $I(1)$. They are cointegrated if there exists β such that

$$y_t - \beta x_t \sim I(0).$$

Economic interpretation

The individual series can drift, but an equilibrium force keeps their long-run relationship from drifting without bound.

Engle-Granger logic: estimate the long-run relation, then test whether its residual is stationary.

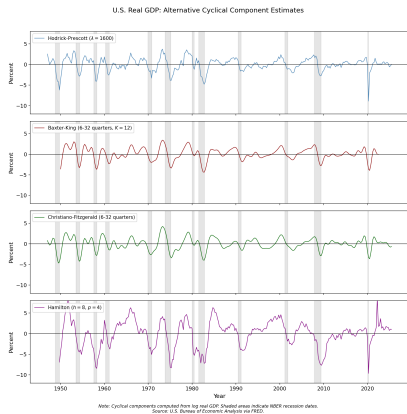
A simple ECM can be written as

$$\Delta y_t = \alpha + \lambda \underbrace{(y_{t-1} - \beta x_{t-1})}_{\text{equilibrium error}} + \gamma \Delta x_t + u_t.$$

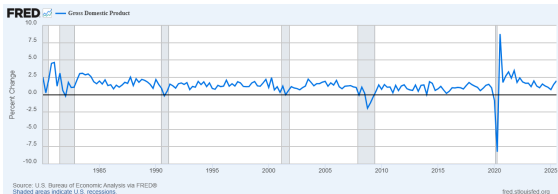
- ▶ γ : short-run response
- ▶ β : long-run relationship
- ▶ λ : speed of adjustment toward equilibrium

Key takeaway

Cointegration links long-run equilibrium with short-run dynamics rather than forcing a choice between levels and differences.



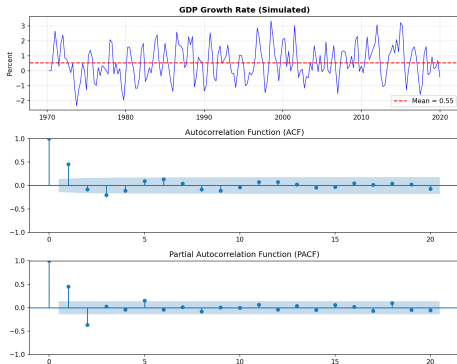
HP, band-pass, Hamilton and unobserved-components approaches encode different assumptions. The extracted “cycle” is therefore model-dependent.



GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

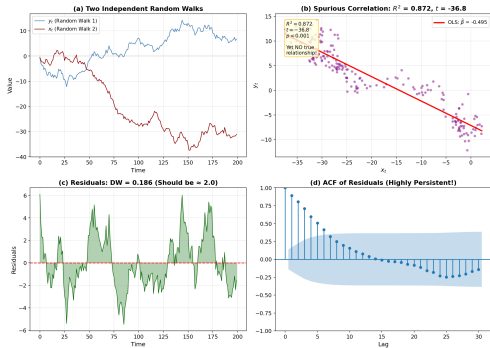
- ▶ α : reaction to recent news
- ▶ β : volatility persistence
- ▶ $\alpha + \beta$: persistence of conditional variance



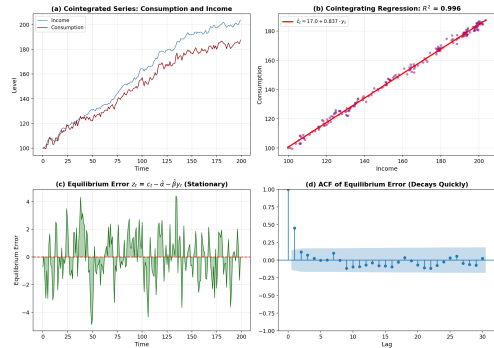
Workflow

1. Identification using time-series behavior and correlograms
2. Estimation of candidate specifications
3. Diagnostic checking of residuals

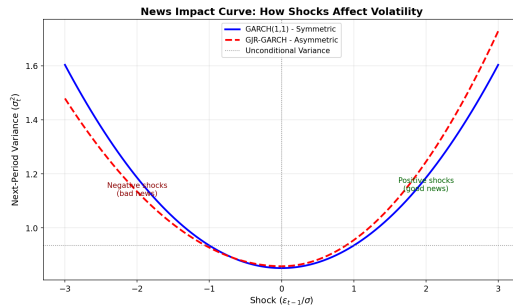
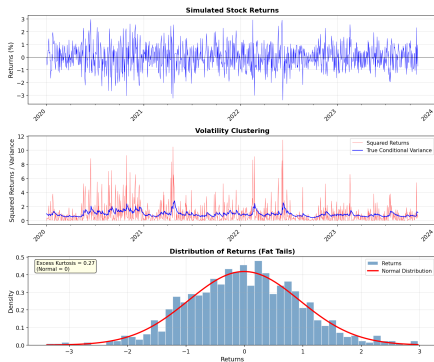
The objective is not merely to minimize an information criterion, but to obtain an adequate dynamic model.



Spurious regression



Cointegration



Volatility clustering motivates conditional variance models; asymmetric variants allow positive and negative shocks to have different effects.

1. **Stationarity comes first**: persistence determines valid modeling strategies.
2. **ACF/PACF are diagnostic tools**, not substitutes for economic judgment.
3. **A unit root changes the meaning of shocks**: effects can become permanent.
4. **Nonstationary regression can be spurious**, unless a genuine cointegrating relation is present.
5. **The conditional mean is only part of the story**: volatility can have its own dynamics.

From one series to a system

Next: Chapter 2 – VAR Models

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